Prepared by group 14

# *Smart Voice Assistant*

Python Project

# *Team Members*


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# *Introduction*

Our project is a python-based **voice assistant** designed to perform tasks using voice commands.

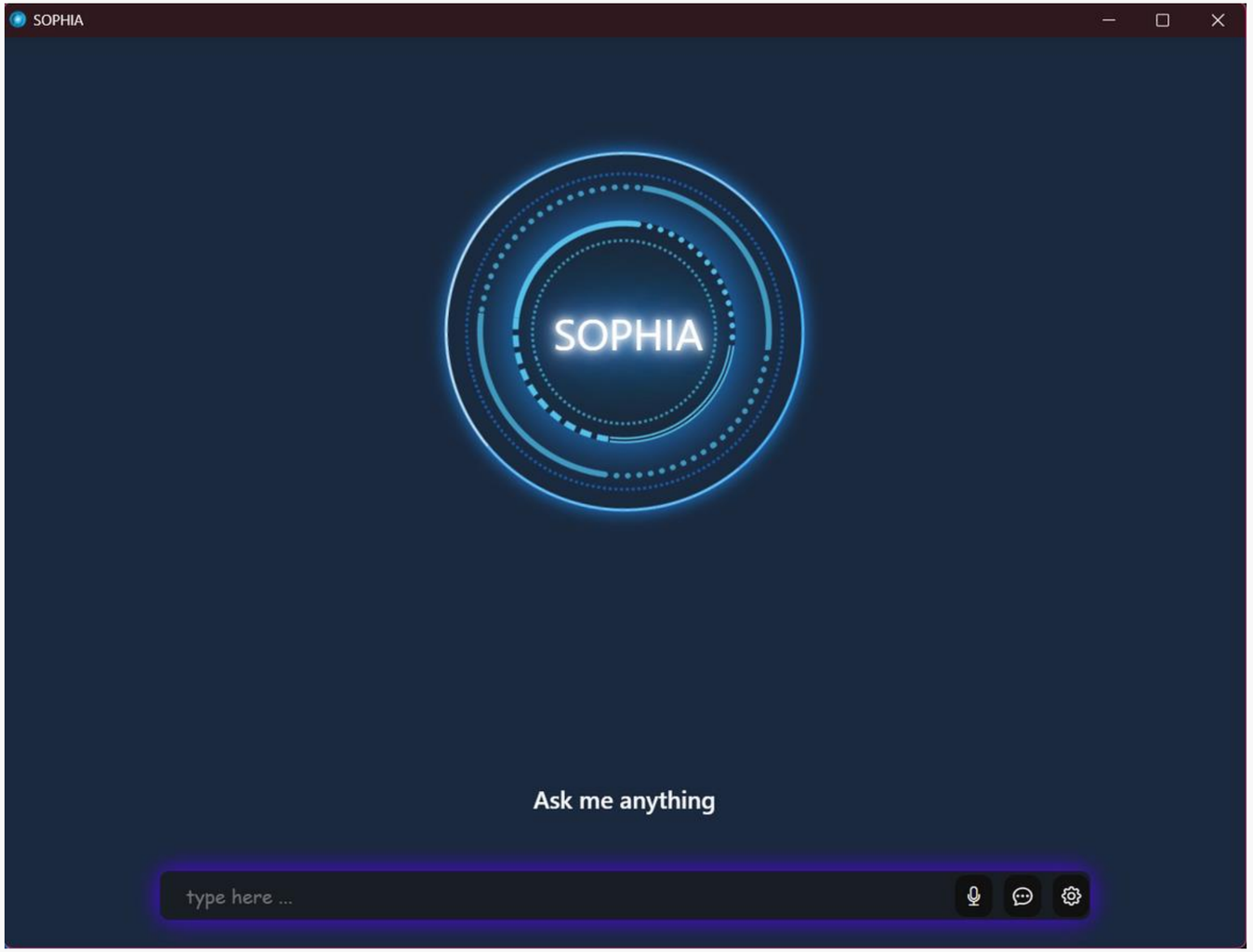
Named **'Sophia'**, the assistant integrates voice recognition, text-to-speech, and automation features.

It is useful for daily productivity and hands-free operation of basic tasks.

# *Technologies Used*

- Python
- Eel (for connecting Python with frontend)
- pyttsx3 (Text-to-Speech)
- SpeechRecognition (Speech-to-Text)
- PyAutoGUI, Requests, Wikipedia, pywhatkit, etc.

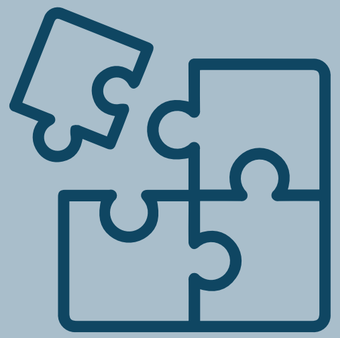
# *Front End*



# *System Architecture*

**Voice Input**

SpeechRecognition captures user input



**Command Processing**

Keywords parsed and directed

**Action Execution**

Features executed via Python modules


**Frontend**

Connected using Eel and displayed in browser

# *Key Features*

- Open applications and websites
- Play YouTube videos
- Fetch latest news and stock prices
- Take notes and screenshots
- Search Google/Wikipedia
- Set alarms and timers

# *Methodology*

**How Commands Work**

- User speaks a command (e.g., 'play music on YouTube')
- Speech is recognized and converted to text
- Text is analyzed to identify the required action
- Corresponding function is called (e.g., PlayYoutube)

# Interesting Code Logic

- Dynamic command routing in command.py
- Threading used for alarms without blocking UI
- Fallback logic for app/URL handling in openCommand()
- Error handling in Wikipedia and News fetching

# *Frontend Integration*

- Eel used to bind Python backend with a web frontend
- Voice responses and messages displayed via HTML/JS
- Browser launched as a desktop app using Chrome app mode

# *EXAMPLE USECASE*

- User says: 'Set a timer for 2 minutes'
- Assistant processes and waits 2 minutes
- Then says: 'Time's up!' via speaker and browser UI

# *FUTURE SCOPE*

- **Better understanding:** Improve how the assistant understands what we say by using smarter language tools.
- **Voice lock:** Add a feature to recognize the user's voice for security.
- **Always listening mode:** Make it listen for a wake word like "Hey Assistant" without clicking a button.

# *Conclusion*

In this project, we built a voice assistant that can understand our voice, perform different tasks like opening apps, searching online, reading news, setting alarms, and more. It talks back to us using text-to-speech and shows messages on a simple interface. The assistant works well, and the project shows how we can use Python to control a computer using voice commands.

*Thank you*